COMP447 · PROGRESS REPORT

When Does Cheap Consistency Tuning Pay Off?

ECT vs Heun under matched latency, and a principled schedule for few step sampling.

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The question

- Diffusion models reach state of the art image quality at the cost of 30 to 100+ neural function evaluations per image.
- Consistency Models compress sampling into 1 or 2 steps, but the upfront tuning is non trivial.
- When is cheap consistency tuning worth paying for, and how should you sample once you've paid?

Background

Diffusion: probability flow ODE [Karras+ 2022, EDM]

$$\frac{dx_t}{dt} = -t\nabla_{x_t}\log p_t(x_t)$$

Consistency model: a learnable flow map [Song+ 2023]

$$f_\theta(x_t, t) \approx x_0, \qquad f_\theta(x_t, t) = f_\theta(x_s, s) \text{ for } (x_t, x_s) \text{ on the same trajectory}$$

- Diffusion: 2 NFE / step, 18 steps ≈ 35 NFE. · CM: 1 NFE for 1 step, 2 NFE for 2 step.

Easy Consistency Tuning · the model we use

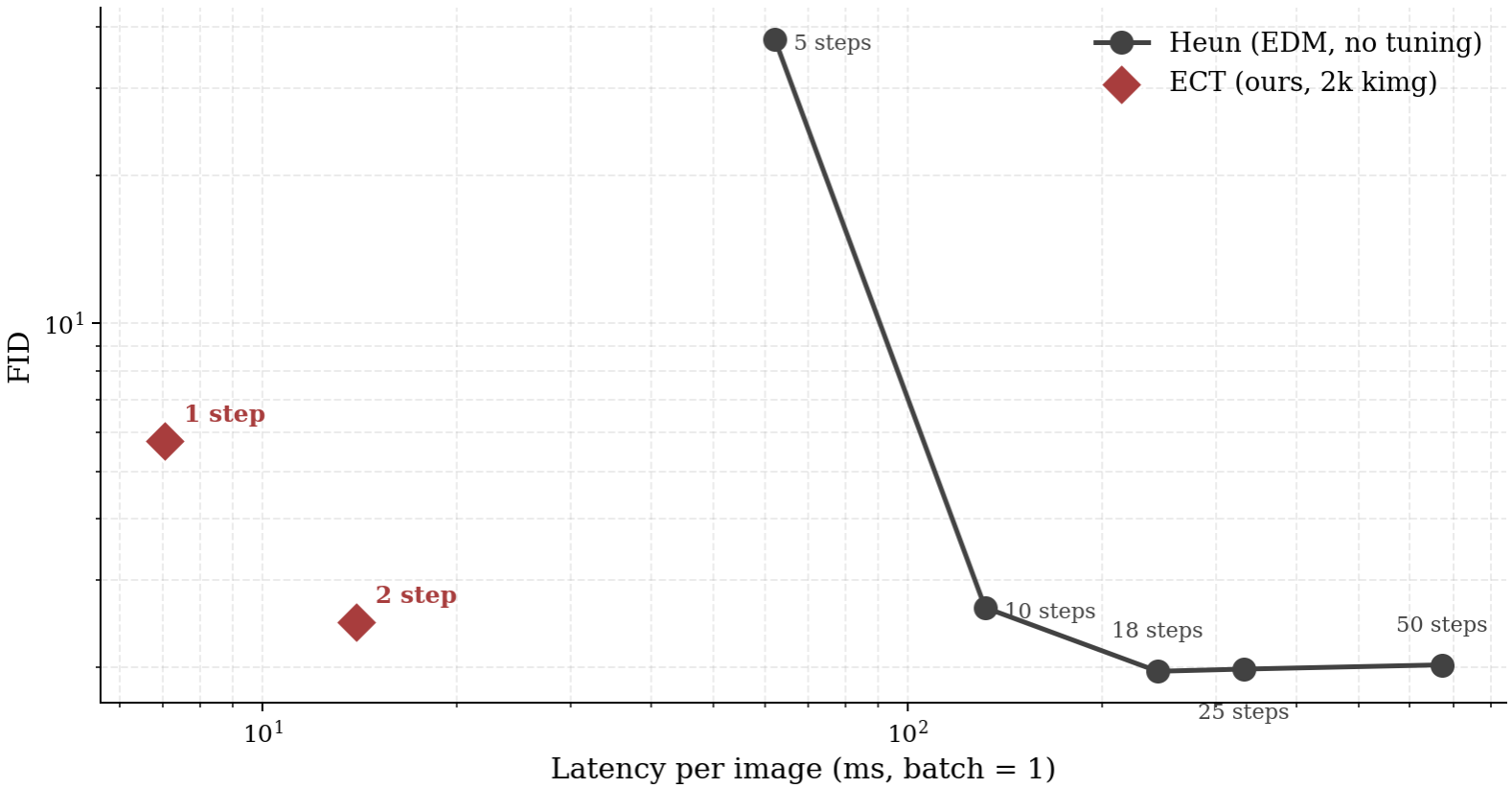
Self consistency between infinitesimally close noise levels [Geng+ 2025]

$$\mathcal{L}_{\mathrm{ECT}}(\theta) = \mathbb{E}_{y,\epsilon,t}\left[\, w(t)\, d\left(f_\theta(y + t\epsilon,\, t),\; f_{\theta'}(y + r\epsilon,\, r)\right)\right], \quad r = t - \Delta(t),\; \Delta(t) \to 0$$

- Initialised from a pretrained EDM checkpoint, no distillation teacher.
- Two step inference uses a single hand picked midpoint at value 0.821, with no derivation in the paper.
- Paper claim: ~1 A100 hour to reach FID 2.11 at 400 k kimg.

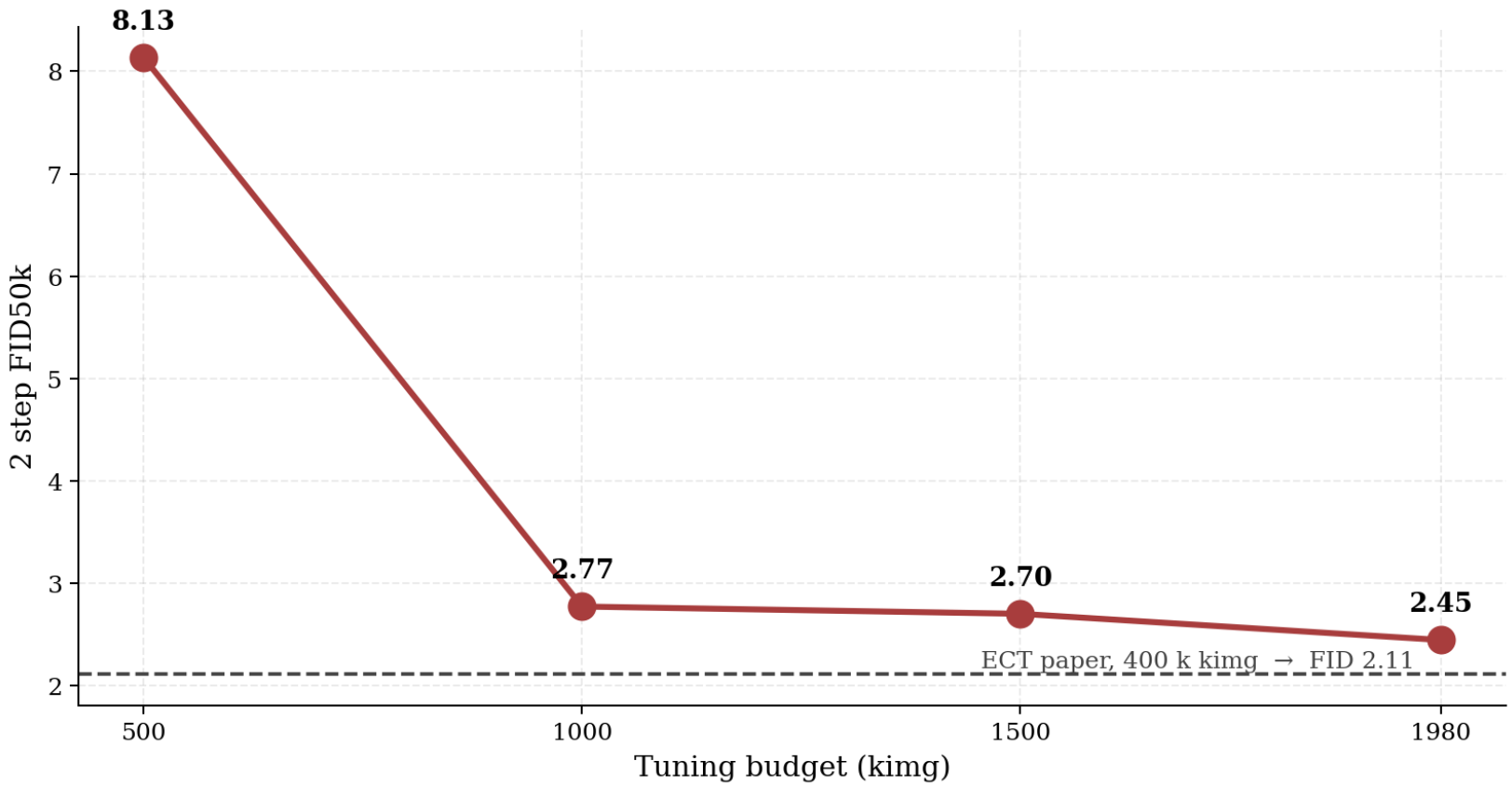
Pareto frontier

EDM CIFAR 10 32×32 · ECT @ 1980 kimg · Heun 5 to 50 steps · NVIDIA G4 Blackwell · batch = 1



- ECT 2 step · FID 2.47 · 14 ms / image → 17× faster than Heun 18.
- On this short tuning budget, ECT Pareto dominates Heun across every operating point.

Tuning budget ablation



- 94 % of paper reported quality at 0.5 % of the paper's training budget.
- Diminishing returns past 1000 kimg → early stopping recipe is realistic.
- Break even vs Heun 18 · $N^* \approx 23.5$ k images (batch 1) before tuning amortises.

The real open problem

- As we characterised 2 step behaviour across our 4 checkpoints, every result rested on a single hidden parameter: a midpoint at 0.821, fixed across all training stages and step budgets.

> “It’s a very good question and I think also an open research problem. I don’t have a manual for it. You can treat it as an optimization problem, maximizing sample quality with respect to these sampling schedules.”
>
> Zhengyang Geng, ECT first author

- No derivation in ECT [Geng+ 2025], sCM [Karras+ 2024], or Multistep CM [Heek+ 2024].
- Heuristic timestep selection exists [TDD, Wang+ 2024], but principled KLUB-based optimization for consistency models does not.
- Align Your Steps [Sabour+ 2024] solves it for diffusion only — we adapt the KL upper bound to the consistency flow map.

Our method · KLUB CM schedule

Re-derive the Align Your Steps KL upper bound for the consistency function, not the score field.

$$\mathcal{T}^* = \arg\min_{\{t_i\}} \sum_{i=1}^{N} \mathbb{E}_{x\sim p}\left[\, \| f_\theta(x_{t_i}, t_i) - f_\theta(x_{t_{i-1}}, t_{i-1}) \|^2 \,\right]$$

Inspiration: KL upper bound [Sabour+ 2024] · trajectory awareness [Kim+ 2024, CTM] · adaptive schedules [OSS, 2025]

- Linearise the flow map (not the score) at chosen knots; the KLUB upper bounds the discretisation error.
- Schedule is checkpoint conditioned → drifts with model quality, unlike a fixed magic number.
- Predicted impact: pushes ECT operating points further left on the Pareto frontier; generalises to unseen step budgets without retraining.

Implementation · what we are coding

- KLUB derivation for the CM flow map · rederived for the consistency function, not the score field.
- Schedule optimiser · LBFGS over the schedule knots in PyTorch — gradient based, orders of magnitude cheaper than grid search.
- On policy loss patch in training/loss.py · student's own outputs at the chosen midpoint, motivated by exposure bias work [Agarwal+ 2023, GKD].
- Corrected evaluation pipeline · 1 step and 2 step routed to the right sampler.

training/loss.py · on policy schedule aware term (sketch)

```
x_mid = f_theta(z * t_max, t_max).detach()         # student's own coarse output
t_mid = klub_optimiser(checkpoint=ckpt)            # KLUB derived knot
L_on  = w(t_mid) * d(
    f_theta      (x_mid + eps * t_mid, t_mid),     # on policy input
    f_theta_prime(x_mid + eps * r,     r),         # EMA target
)
L     = L_ECT  +  lam * L_on                       # original + on policy
```

Status & next steps

Done

- ECT tuning, 4 checkpoints saved.
- Heun baselines @ 5 to 50 steps.
- Latency profiling, B = 1 and B = 64.
- Corrected evaluation pipeline.

In flight · Up next

- Midpoint grid sweep across all 4 checkpoints.
- KLUB CM derivation + schedule optimiser.
- On policy schedule aware fine tune (stretch).
- Final report · early June.

Risk · GPU time on Colab; mitigated by small checkpoint footprint and 10k sample screening for the sweep.